Hi team, here is a quick overview of the **SunTax** platform for your testing and review. This guide covers where the application is hosted, the test accounts you can use, a simple 5-step walkthrough to test the full declaration flow, and our recommendations for production hosting, storage, and Twilio subscriptions.

1. Live Deployment & Repository Links

Component	URL / Location	Notes
Web App (Frontend)	**https://frontend-ruby-gamma-46.vercel.app/**	Hosted on Vercel. Fully responsive across desktop and mobile.
Backend API	**https://suntax-api.onrender.com**	FastAPI service handling calculations, OCR, and user sessions.
API Docs (Swagger)	**https://suntax-api.onrender.com/api/docs**	Interactive API explorer to inspect all endpoints and test requests.
GitHub Repository	**https://github.com/usamashaikh13/suntax**	Full codebase: Next.js frontend, Python backend, and Swiss tax rules.

Note on initial loading: The backend is currently running on Render's free tier. If the server has been idle, the very first request takes roughly 45–60 seconds to spin up. All subsequent clicks and API calls respond immediately.

2. Demo Accounts & Logins

Account	Credentials	2FA Setup & Access
Primary Test Account Full 2FA enabled	**Email:** client@suntax.ch **Password:** TestPassword123!	Requires a 6-digit 2FA code. • **Authenticator Key:** OWZDANSQXHFBZXANGHT3ZKUYM2XW2AAV • Or run: python get_2fa.py client@suntax.ch
Standard Demo Account Direct sign-in	**Email:** demo@suntax.ch **Password:** TestPassword123!	Logs in directly without 2FA prompt. You can test setting up 2FA on your phone from *My Account → Security*.

3. How to Test the Application (5-Step Walkthrough)

1. Sign In	Go to the live web app and log in with **client@suntax.ch** / **TestPassword123!**. Enter the 6-digit code from your authenticator app when prompted.
2. Start Return	On your dashboard, click **+ New Tax Return**. Select Canton **ZH (Zürich)**, Municipality **Zürich (BFS #261)**, and Tax Year **2025**, then click confirm.
3. Upload Documents	In the **Documents** tab, drag and drop test documents (e.g. Salary Certificate, Bank Statement, Pillar 3a Certificate, or Donation Receipt). The OCR worker extracts financial data automatically.
4. Review Extracted Data	Click **Review Fields** on any processed document to verify the extracted figures, then click **Approve & Apply to Profile** to save them to your tax declaration.
5. Calculate & Submit	Go to the **Calculation** tab to run the official Swiss tax engine (Federal, Cantonal, and Municipal breakdown). When ready, proceed to **Review & Submit** and finalize the declaration.

4. Database & Codebase Details

• **Local SQLite Database:** Located at backend/suntax_dev.db. Pre-seeded with tax rules for test cantons, demo accounts, and sample declarations for immediate local development.

• **Production PostgreSQL 16:** Uses asyncpg with Row-Level Security (RLS) policies. Each user's financial documents and declarations are isolated strictly at the database layer.

• **Document File Storage:** Uploaded files are stored in backend/dev_storage/ during development (mirroring S3 bucket structure with user-prefixed paths). In production, this maps to an S3 bucket.

• **Running Locally:** Run ./backend/start_dev.sh to create the Python virtualenv and start the backend, and npm run dev inside frontend/ to run the web interface.

5. Core API Endpoints Reference

Endpoint	Method	Purpose
`/api/v1/auth/login`	POST	User authentication; returns JWT token or 2FA challenge
`/api/v1/auth/login/2fa`	POST	Validates 6-digit TOTP code and returns full session token
`/api/v1/tax-returns`	GET / POST	Lists all user tax returns or creates a new draft declaration
`/api/v1/documents/upload`	POST	Uploads tax documents, generates hash, and triggers OCR pipeline
`/api/v1/tax-returns/{id}/calculate`	POST	Runs deterministic Swiss tax calculation for federal, canton & city
`/api/v1/tax-returns/{id}/confirm`	POST	Finalizes the tax declaration with 2FA verification
`/api/v1/tax-returns/{id}/export/pdf`	POST	Generates official tax declaration summary PDF for download

6. Production Hosting, Storage & Twilio Subscriptions

For production rollout, infrastructure must comply with the **Swiss Federal Act on Data Protection (FADP / nDSG)**. Below are the recommended tiers for compute, sovereign Swiss storage, and communication services:

A. Document Vault & Database Hosting (100% Swiss Soil)

• **Storage — Infomaniak Swiss S3 (Geneva / Winterthur):** 100% Swiss jurisdiction (no US Cloud Act). ISO 27001 data centers, CHF 0.012/GB/mo, zero egress fees.

• **Backend & Worker — Exoscale Cloud VPS (Zürich, ~CHF 38/mo):** Runs FastAPI & OCR in Zürich on high-performance Swiss hardware.

• **Database — Exoscale Managed PostgreSQL 16 (Zürich, ~CHF 45/mo):** Automated daily backups, PITR, and encryption at rest in Zürich.

• **Frontend — Vercel Pro (~$20/mo):** Global AnyCast CDN with sub-30ms response times for Swiss users and zero maintenance.

B. Communication Providers: Twilio Subscriptions (Email & Phone SMS)

• **Email Service — Twilio SendGrid (Essentials Plan, ~$19.95 / mo):** Production transactional email engine for verification links, password resets, declaration ready notices, and summary PDF delivery. Includes high inbox deliverability and DKIM/SPF domain authentication.

• **Phone SMS Provider — Twilio SMS API (~$10–15 / mo):** Sends instant 2FA login verification codes and filing status updates directly to Swiss mobile numbers (+41). Billed pay-as-you-go (~CHF 0.04–0.08 per SMS).

C. Production Monthly Budget (Swiss Cloud + Twilio Stack)

Service Component	Provider & Subscription Tier	Est. Cost
Frontend Web App	Vercel Pro (Global Edge)	~$20 / mo
Backend API & Worker	Exoscale Cloud VPS (Zürich, Switzerland)	~CHF 38 / mo
Managed Database	Exoscale Managed PostgreSQL 16 (Zürich)	~CHF 45 / mo
Secure Document Vault	Infomaniak Swiss S3 (Geneva / Winterthur)	~CHF 5–10 / mo
Transactional Email	Twilio SendGrid (Essentials Plan)	~$20 / mo
Phone SMS Verification	Twilio SMS API (Pay-as-you-go for Swiss +41)	~$10–15 / mo
Total Monthly Cost	**Complete Production Stack (Swiss Cloud + Twilio)**	**~CHF 140–155 / mo**